



Second Derivatives Using Implicit Differentiation

Calculus Worksheet · Grade 11-12

Name: _____

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Learning Objectives

- Apply implicit differentiation to find the first derivative dy/dx of equations relating x and y
- Use the quotient rule to differentiate rational expressions involving y
- Simplify complex fractions to obtain a clean form of the second derivative

Find the first and second derivatives of each implicit equation, then simplify your final answer using algebra.

1. Find the second derivative of the following implicit equation.

$$x^2 + y^2 = 25$$

Answer: _____

2. Find the first derivative dy/dx by implicit differentiation.

$$x^2 + y^2 = 49$$

Answer: _____

3. Find the second derivative of this circle equation.

$$x^2 + y^2 = 16$$

Answer: _____

4. Use implicit differentiation to find dy/dx .

$$x^2 - y^2 = 9$$

Answer: _____

5. Find the second derivative for the given equation.

$$x^2 + 4y^2 = 36$$

Answer: _____

6. Find dy/dx for the given relation.

$$xy = 12$$

Answer: _____

7. Find the second derivative of the relation.

$$xy = 10$$

Answer: _____



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8. Find dy/dx by implicit differentiation.

$$x^3 + y^3 = 8$$

Answer: _____

9. Find the second derivative for the implicit equation.

$$x^2 + y^2 = 1$$

Answer: _____

10. Find dy/dx for the equation below.

$$x^2 + xy + y^2 = 7$$

Answer: _____





Remind students to substitute dy/dx back into the second derivative and to use the original equation to simplify when possible.

Solutions

1. Find the second derivative of the following implicit equation.

$$x^2 + y^2 = 25$$

- Differentiate both sides implicitly to get $2x$ plus $2y$ times dy/dx equals zero.
- Solve for dy/dx to obtain negative x over y .
- Apply the quotient rule to negative x over y to get the second derivative.
- Substitute dy/dx equals negative x over y and simplify using x squared plus y squared equals 25.

Answer:
$$\frac{d^2y}{dx^2} = -\frac{25}{y^3}$$

2. Find the first derivative dy/dx by implicit differentiation.

$$x^2 + y^2 = 49$$

- Differentiate each term with respect to x .
- The derivative of y squared is $2y$ times dy/dx .
- Isolate dy/dx by dividing through.

Answer:
$$\frac{dy}{dx} = -\frac{x}{y}$$

3. Find the second derivative of this circle equation.

$$x^2 + y^2 = 16$$

- Find the first derivative using implicit differentiation to get negative x over y .
- Apply the quotient rule to differentiate negative x over y .
- Substitute the first derivative back in and combine fractions.
- Use the original equation to express the result in terms of y only.

Answer:
$$\frac{d^2y}{dx^2} = -\frac{16}{y^3}$$

4. Use implicit differentiation to find dy/dx .

$$x^2 - y^2 = 9$$

- Differentiate both sides with respect to x .
- The derivative of negative y squared is negative $2y$ times dy/dx .
- Solve algebraically for dy/dx .

Answer:
$$\frac{dy}{dx} = \frac{x}{y}$$



5. Find the second derivative for the given equation.

$$x^2 + 4y^2 = 36$$

- Differentiate to find dy/dx equals negative x over four y .
- Apply the quotient rule to that expression.
- Substitute dy/dx back into the formula.
- Use the original equation to simplify the numerator.

Answer: $\frac{d^2y}{dx^2} = -\frac{9}{4y^3}$

6. Find dy/dx for the given relation.

$$xy = 12$$

- Differentiate using the product rule.
- You get y plus x times dy/dx equals zero.
- Isolate dy/dx by dividing.

Answer: $\frac{dy}{dx} = -\frac{y}{x}$

7. Find the second derivative of the relation.

$$xy = 10$$

- Differentiate implicitly to get dy/dx equals negative y over x .
- Apply the quotient rule to differentiate negative y over x .
- Substitute the first derivative into the resulting expression.
- Combine and simplify the fraction.

Answer: $\frac{d^2y}{dx^2} = \frac{2y}{x^2}$

8. Find dy/dx by implicit differentiation.

$$x^3 + y^3 = 8$$

- Differentiate both sides to get three x squared plus three y squared times dy/dx equals zero.
- Divide both sides by three.
- Solve for dy/dx .

Answer: $\frac{dy}{dx} = -\frac{x^2}{y^2}$

9. Find the second derivative for the implicit equation.

$$x^2 + y^2 = 1$$

- Differentiate to obtain dy/dx equals negative x over y .
- Apply the quotient rule to negative x over y .
- Substitute dy/dx back in and simplify the complex fraction.
- Use the identity x squared plus y squared equals one to simplify.

Answer: $\frac{d^2y}{dx^2} = -\frac{1}{y^3}$



10. Find dy/dx for the equation below.

$$x^2 + xy + y^2 = 7$$

→ Differentiate each term, using the product rule on xy .

→ Group all dy/dx terms on one side.

→ Factor and solve for dy/dx .

Answer: $\frac{dy}{dx} = -\frac{2x + y}{x + 2y}$

